Gold Price Prediction using Machine Learning

A Mini Project in Machine Learning

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Introduction

- Gold is a globally traded precious metal and a traditional safe-haven asset.
- Its price is affected by demand, inflation, currency value and market sentiment.
- Being able to forecast gold prices helps investors and analysts take better decisions.
- This project builds a Machine Learning model that predicts the monthly price of gold.

Problem Statement

- Given the historical monthly price of gold, predict its price for the coming months.
- This is a supervised regression problem: past behaviour is the input, future price is the output.
- Multiple models are trained and the most accurate one is kept for prediction.
- The winning model is then wrapped in a simple desktop application.

Objectives

- Prepare a clean monthly gold price dataset for the years 2000 to 2024.
- Engineer features that capture the trend and the short-term momentum of prices.
- Train and compare RandomForest, Linear Regression and Ridge models.
- Measure accuracy using MAE, RMSE and R-squared against a naive baseline.
- Provide a simple desktop GUI so anyone can run a prediction.

Dataset

- Monthly gold price in USD per troy ounce, from 2000 to 2024 (300 records).
- Columns in the file: Date, Year, Month and Price.
- Built from public annual-average anchors with a small momentum effect added.
- Model features: the last 3 monthly returns, plus the month and the year.

Methodology

- The model predicts the month-to-month RETURN, not the raw price.
- Why: tree models cannot guess a price higher than they have seen, so a rising trend breaks them.
- Price is rebuilt as next price = last price x (1 + predicted return).
- Forecasting rolls forward one month at a time, feeding each answer back as the next input.
- A chronological split is used - the last 60 months are kept for testing.

Technology Used

- Language: Python.
- Data handling: pandas and NumPy.
- Models: scikit-learn - RandomForest, Linear Regression, Ridge.
- Charts: Matplotlib.
- Interface: Tkinter. Save / Load: Joblib.

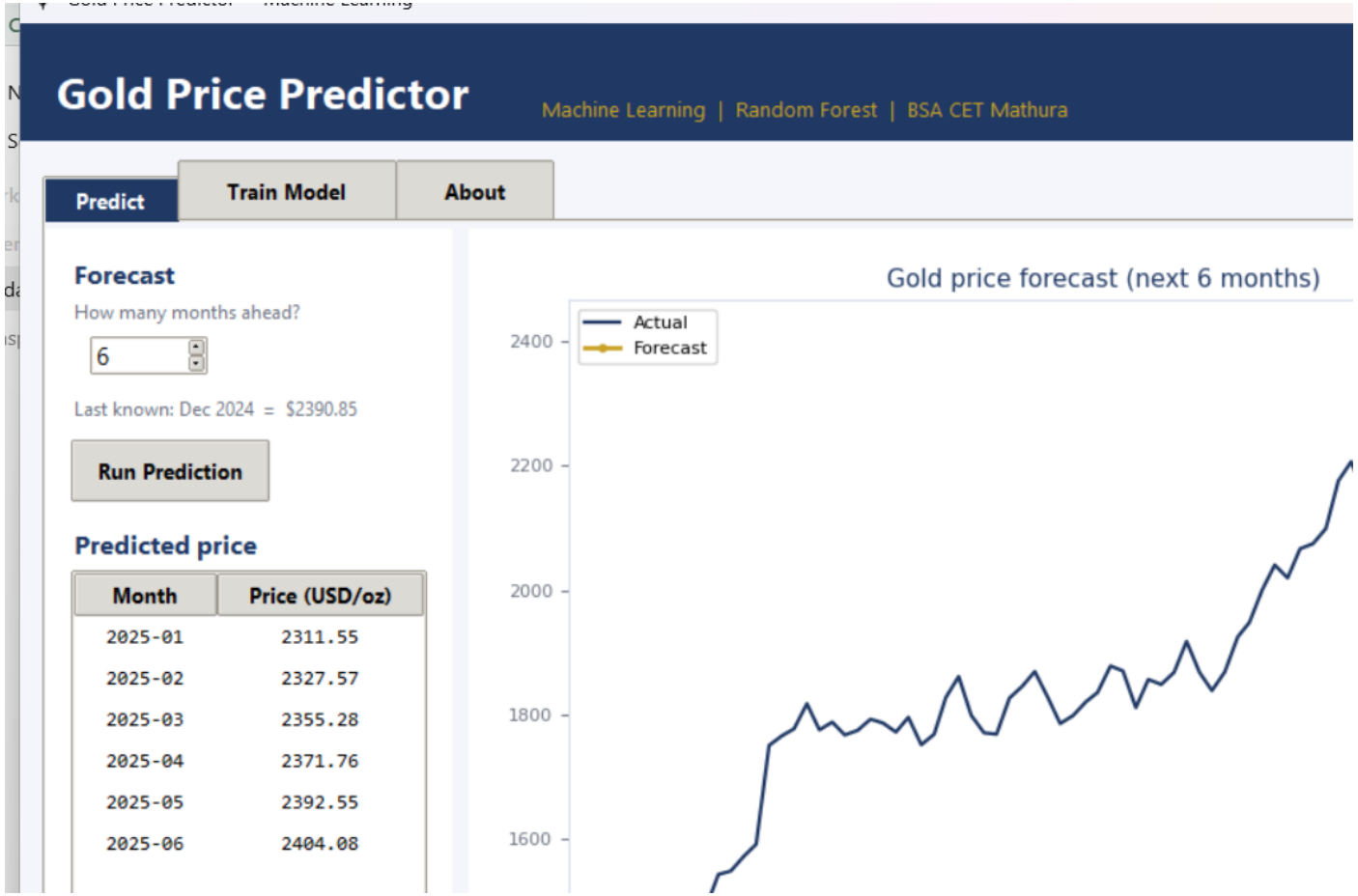
Results

Model	MAE (USD)	RMSE (USD)	R-squared
RandomForest	32.71	44.47	0.9613
Linear Regression	32.91	44.60	0.9611
Ridge	33.33	45.50	0.9595
Naive baseline	36.14	47.97	0.9549

RandomForest wins on every metric. Test window: last 60 months (from Jan 2020). Lower MAE / RMSE and higher R-squared are better.

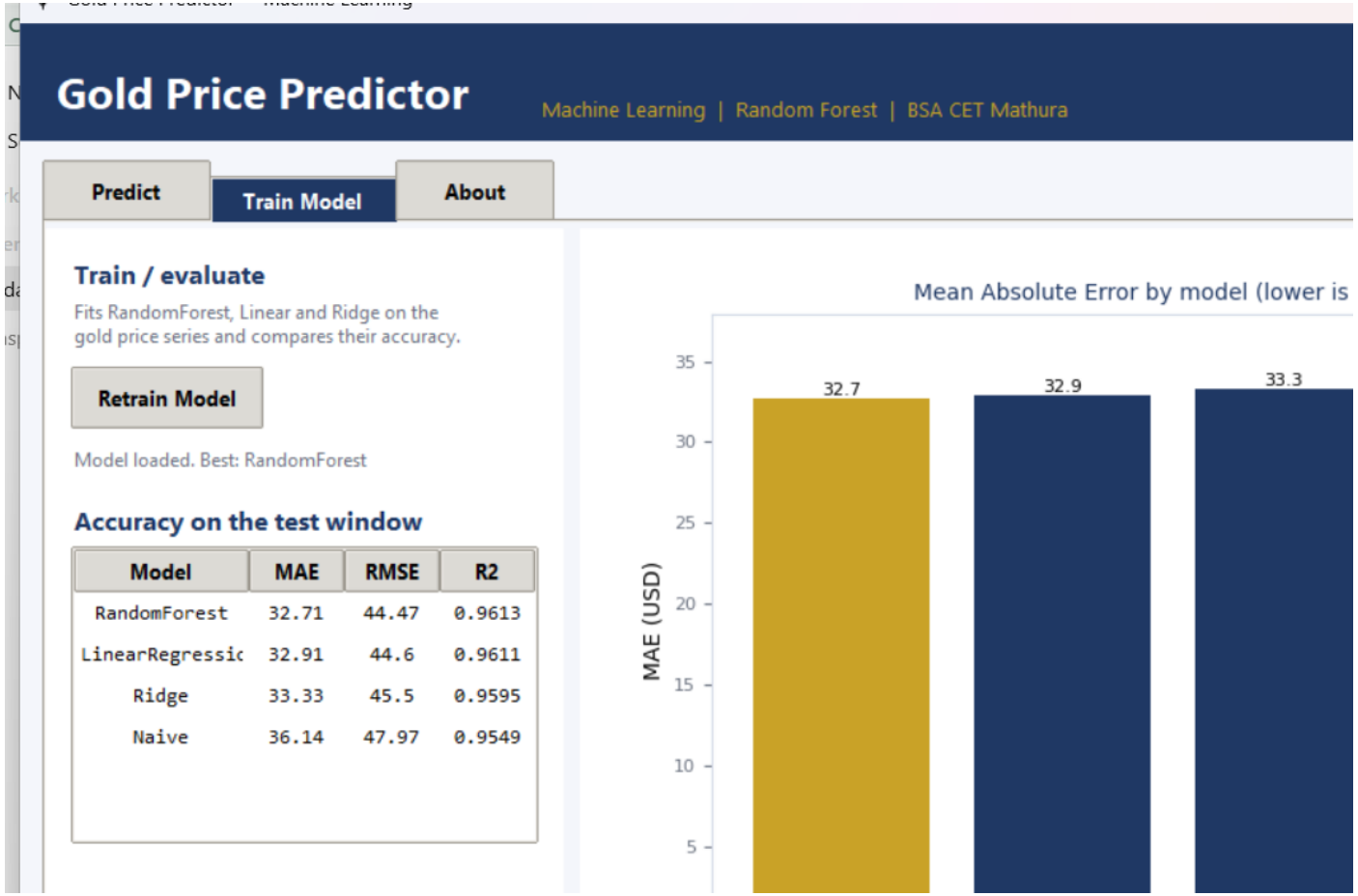


Application - Prediction



Desktop GUI: choose a horizon and forecast the gold price month by month.

Application - Model Comparison



Train tab: accuracy table and the error comparison across all models.

Conclusion

- RandomForest gave the best accuracy and clearly beat the naive baseline.
- Predicting returns instead of raw prices made the models reliable on a trending series.
- The Tkinter GUI turns the trained model into an easy-to-use forecasting tool.
- The project demonstrates a complete Machine Learning pipeline from data to interface.

Future Scope

- Add real economic features such as inflation, USD index and interest rates.
- Try deep-learning time-series models like LSTM for longer forecasts.
- Connect a live gold price API so predictions always use current data.
- Show a confidence range instead of a single predicted value.



Thank You

Questions & suggestions are welcome

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